6.1 Lines

Name _____ Period ____ Date _____

Pre-Calculus 6.1 (30-minute plan) · Student sheet

I can…

- find the inclination of a line from its slope or equation.
- find the angle between two lines.
- find the distance between a point and a line.

Vocabulary · fill in the missing word

Inclination — the _____ θ ($0 \le \theta < \pi$) measured counterclockwise from the positive x-axis to the line

Angle between two lines — the _____ of the angles the lines form, $0 < \theta \le \pi/2$

Part 1 · Notes and examples (fill in each note, then work the example)

Inclination and slope: m = _____

Example 1

Find the inclination of $x + y = 2$.

Angle between two lines: $\tan\theta$ = _____

Example 2

Find the angle between $2x - y = 4$ and $3x + 4y = 12$. Round to the nearest tenth of a degree.

Distance from (x_1, y_1) to $Ax + By + C = 0$: d = _____

Example 3

Find the distance from $(4, 1)$ to $y = 2x + 1$.

Part 2 · Partner practice

Partner practice

Find the distance from $(2, -1)$ to $3x + 4y - 5 = 0$.